Milestone 2 :

MODEL TRAINING & EVALUATION

Project: ExoHabitAI–PredictingHabitabilityofExoplanetsusing Machine Learning

1. Objective

The objective of the model training phase is to develop a machine learning classifier capable of predicting whether an exoplanet has the potential to support habitable conditions based on its planetary and stellar characteristics. Instead of defining strict physical rules manually, the machine learning model learns patterns from historical exoplanet observations obtained from astronomical datasets such as the NASA Exoplanet Archive.

The model predicts one of two classes:

CLASS	MEANING
1 — Potentially Habitable	Planet satisfies basic physical conditions for habitability
0 — Non-Habitable	Planet conditions unsuitable for habitability

The dataset initially contained thousands of observations with numerous planetary and stellar parameters. After preprocessing and feature engineering, a smaller set of relevant features was used for training the models.

2. Feature and Target Separation

Before training the machine learning models, the dataset was separated into:

Inputs

Cleaned planetary and stellar features such as:

- Planet radius (pl_rade)

- Orbital distance (pl_orbsmax)
- Orbital period (pl_orbper)
- Equilibrium temperature (pl_eqt)
- Stellar luminosity (st_lum)
- Stellar temperature (st_teff)
- **Engineered features such as:**
    - habitability_score
    - orbital_stability
    - stellar_compatibility

Output

- Habitability class
- Habitability probability score

Dataset Preparation

The machine learning dataset was created after data cleaning and feature engineering.

Input Dataset

```
data/processed/exohabit_ml.csv
```

Preparation Steps

Selected relevantplanetary and stellar features.

1.

2.Removed irrelevant and high-missing columns.

3.Handled missing values using median imputation.

4.Removed invalid values such as negative radius or orbital parameters.

5.Applied outlier removal using IQR method.

6. Engineered additional features including:

- habitability_score
- stellar_compatibility

- orbital_stability

3. Train-Test Split

DATASET	PURPOSE	SIZE
TRAINING SET	Used for model learning	80%
TESTING SET	Used for model evaluation	20%

This ensures unbiased model evaluation.

Random state was fixed for reproducibility.

4. Baseline Models

Baseline models were used to establish an initial benchmark before applying more advanced algorithms.

MODEL	KEY CHARACTERISTIC	REASON FOR SELECTION
LOGISTIC REGRESSION	Linear classifier	Serves as a baseline for comparative analysis
DECISION TREE	Rule-based splits	Offers an interpretable model
RANDOM FOREST	Ensemble learning	Demonstrates robust performance
XGBOOST	Gradient boosting	Exhibits exceptional performance on tabular data

To ensure proper preprocessing and prevent data leakage, a machine learning pipeline was created. The pipeline combines:

- Feature scaling
- Model training

Example pipeline implementation:

```
pipeline = Pipeline([ ("scaler", StandardScaler()), ("model", RandomForestClassifier( n_estimators=200, random_state=42 )) ])
```

Benefits of pipelines:

- Prevents data leakage
- Ensures consistent preprocessing
- Simplifies model training workflow

6. Model Training

The selected models were trained using the training dataset.

Default parameters were used initially, followed by hyperparameter tuning.

The command to fit the pipeline is as follows: pipeline.fit(X_train, y_train) This instructs the pipeline to learn from the training data, X_train and the corresponding labels, y_train.

During training:

1. Features are standardized using StandardScaler.

2.The model learns patterns from the training data.

3.Internal model parameters are optimized to minimize prediction error.

The trained model is then used to generate predictions for the testing dataset.

Model performance was evaluated using multiple classification metrics.

METRIC	PURPOSE
ACCURACY	Overall correctness
PRECISION	Reliability of habitable predictions
RECALL	Ability to detect habitable planets
F1-SCORE	Balance of precision and recall
ROC-AUC	Model discrimination capability

7.1 Evaluation Results

MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
LOGISTIC RE-GRESSION	0.996	1	0.99	0.99
RANDOM FOREST	1	1	1	1
XGBOOST	1	1	1	1

From the evaluation results:

Random Forest and XGBoost achieved perfect classification performance on the testing dataset. Logistic Regression also performed extremely well, achieving nearly perfect accuracy. The ensemble models (Random Forest and XGBoost) slightly outperform Logistic Regression due to their ability to capture complex nonlinear relationships. These results indicate that the dataset contains strong patterns allowing models to clearly separate habitable and non-habitable planets.

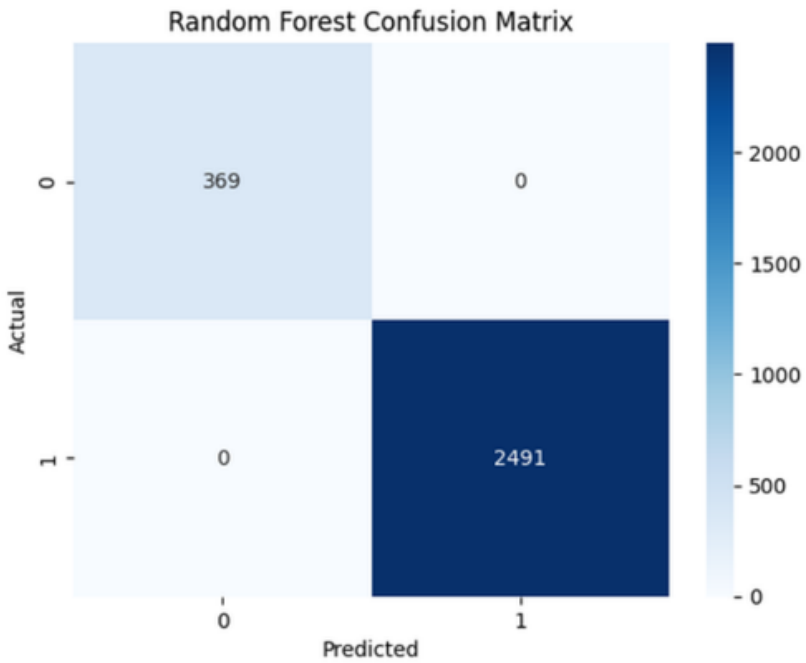
7.2 Confusion Matrix Analysis

Theconfusion matrix provides adetailed view of how the model's predictions compare with the actual class labels. It shows the number of correct and incorrect predictions for each class.

For the Random Forest and XGBoost models, the confusion matrix indicates perfect classification performance on the testing dataset.

Confusion Matrix Structure

ACTUAL / PREDICTED	PREDICTED 0 (NON-HABITABLE)	PREDICTED 1 (HABITABLE)
ACTUAL 0	True Negative (TN)	False Positive (FP)
ACTUAL 1	False Negative (FN)	True Positive (TP)



Interpretation

True Negatives (369)

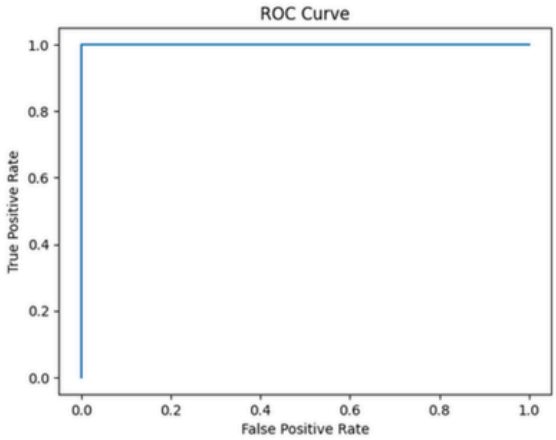
The model correctly identified all non-habitable planets. True Positives (2491) The model correctly classified all habitable planets. False Positives (0) No non-habitable planets were incorrectly classified as habitable.

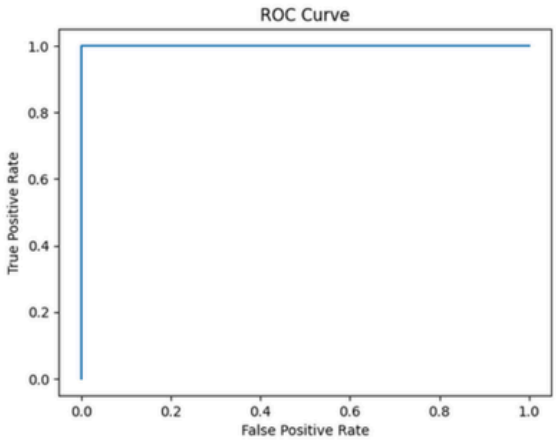
False Negatives (0) No habitable planets were missed by the model. This indicates perfect classification on the test dataset.

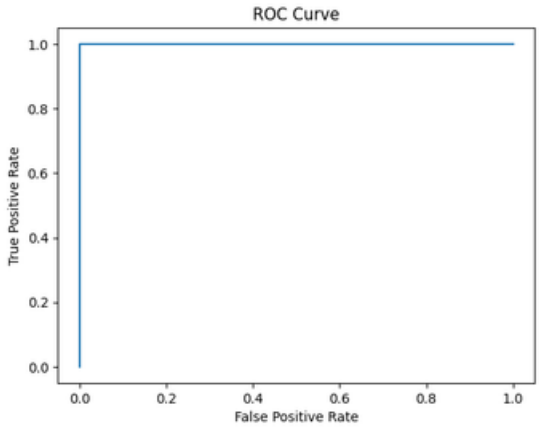
The Receiver Operating Characteristic (ROC) Curve evaluates the ability of a classification model to distinguish between the two classes — habitable and non-habitable exoplanets. The ROC curve plots:

- True Positive Rate (TPR) → Recall
- False Positive Rate (FPR) across different classification thresholds.

A model that perfectly separates both classes will have an ROC curve that rises sharply toward the top-left corner of the graph.







Logistic Regression ROC (AUC = 0.99)

Random Forest ROC (AUC = 1)

XG Boost ROC(AUC=1)

8. Hyperparameter Tuning

Hyperparametertuning is theprocess of optimizing model parameters that control the learning process. These parameters are not learned automatically by the model and must be configured before training. Tuning helps improve model performance by finding the most effective combination of parameter values. In this project, GridSearchCV was used to systematically test different combinations of hyperparameters for the machine learning models.

Random Forest Hyperparameter Tuning: For the Random Forest model, the following parameters were tuned:

PARAMETER	DESCRIPTION
N_ESTIMATORS	Number of decision trees in the forest
MAX_DEPTH	Maximum depth of each tree
MIN_SAMPLES_SPLIT	Minimum samples required to split a node

```
param_grid = { "n_estimators": [100, 200], "max_depth": [5,
10, None] }
grid = GridSearchCV( RandomForestClassifier(random_state=42),
param_grid, cv=3, scoring="f1", n_jobs=-1 )
```

The GridSearch algorithm evaluates each parameter combination using cross-validation and selects the configuration that maximizes the chosen evaluation metric.

9. Feature Importance Analysis

Understanding which variables moststrongly influence the model's predictions is essential for interpreting machine learning results. Feature importance analysis helps identify which planetary and stellar characteristics contribute most to the classification of potentially habitable exoplanets. In this project, feature importance scores were obtained from the Random Forest model, which measures how much each feature reduces prediction error across all decision trees in the ensemble. Features with higher importance values are used more frequently and provide greater predictive value during model decision splits. The analysis revealed that three features play a dominant role in predicting exoplanet habitability.

9.1 Planet Radius (pl_rade)

Importance: Highest Planetradius represents the size of the planet relative to Earth, measured in Earth radii. From an astrophysical perspective, planet size is a crucial indicator of planetary composition:

- Planets with radii between 0.5 and 2.5 Earth radii are typically rocky terrestrial planets. Rocky planets can support solid surfaces, which are essential for maintaining stable atmospheres and potential liquid water.
- Larger planets are usually gas giants, composed primarily of hydrogen and helium, and are unlikely to support Earth-like habitability.

Because of these physical constraints, the model heavily relies on planet radius to distinguish between rocky Earth-like planets and gas giants.
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9.2 Stellar Luminosity (st_lum)

Importance: SecondHighest Stellar luminosity measures the total energy emitted by the host star. This property directly influences the amount of radiation received by orbiting planets. Luminosity plays a key role in determining whether a planet lies within the Habitable Zone (HZ) — the region around a star where liquid water could potentially exist. Important considerations include:

- High luminosity can cause planets to become too hot, leading to atmospheric loss or runaway greenhouse effects.
- Low luminosity may result in extremely cold planetary conditions, preventing liquid water formation.

Therefore, stellar luminosity strongly affects a planet's equilibrium temperature and environmental stability, making it a critical variable for habitability prediction.

9.3 Habitability Score (habitability_score)

Importance: ThirdHighest The habitability scoreis an engineered feature designed to measure how closely a planet resembles Earth-like conditions. This composite metric incorporates deviations from Earth in several parameters:

- Planet equilibrium temperature
- Planet radius
- Orbital distance from the star

- Stellar luminosity

Planets with lower deviations from Earth-like values receive higher habitability scores, indicating conditions that may be more favorable for supporting life. By combining multiple physical factors into a single metric, the habitability score helps the model identify planets with balanced environmental conditions similar to Earth.

9.4 Interpretation of Feature Importance

The feature importance results demonstrate that the machine learning model primarily relies on physically meaningful astrophysical parameters when predicting habitability. The three most influential factors — planet size, stellar radiation, and Earth-like environmental similarity — are widely recognized in planetary science as fundamental determinants of habitability. This alignment between the machine learning model and established astrophysical principles suggests that the model is learning scientifically valid patterns from the dataset, rather than relying on arbitrary correlations.

10. Final Interpretation of Results

After training and evaluating multiple machine learning models, the results demonstrate that the selected features provide strong predictive signals for distinguishing between potentially habitable and non-habitable exoplanets. The models evaluated in this study include:

- Logistic Regression
- Random Forest
- XGBoost

All models achieved extremely high evaluation scores, with Random Forest and XGBoost producing perfect classification metrics on the testing dataset. This indicates that the chosen planetary and stellar features provide clear separation between the two classes.

10.1 Summary of Model Performanceults

The table below summarises the key evaluation metrics for each model.

MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
Logistic Regression	0.996	1	0.99	0.99

MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE
Random Forest	1	1	1	1
XGBoost	1	1	1	1

These results show that ensemble-based algorithms such as Random Forest and XGBoost outperform the baseline Logistic Regression model.

10.2 Model Selection

Based on the evaluation results, XGBoost and Random Forest demonstrated the strongest performance. These algorithms are particularly effective for structured tabular datasets because they can capture complex nonlinear relationships between features. Random Forest was also selected as a reliable model due to its stability, interpretability through feature importance, and robustness against overfitting.

The final trained model was used to generate habitability probabilities for each exoplanet. These probabilities represent the likelihood that a given planet satisfies the conditions required for potential habitability. These probabilities were converted into a habitability score, allowing planets to be ranked from most to least habitable.
This dataset contains:

- Planetary features
- Habitability score
- Planet ranking

This ranking provides a useful tool for identifying promising exoplanet candidates for further astronomical study.

10.4 Key Insights

Several important insights emerged from the analysis:

1. Planet radius is the strongest predictor of habitability, highlighting the importance of distinguishing rocky planets from gas giants.

2. Stellar luminosity significantly influences planetary temperature, which determines whether liquid water could exist on the planet's surface.

3. Engineered habitability metrics improve prediction accuracy by combining multiple astrophysical factors into a single interpretable feature.

These findings align closely with established scientific theories regarding planetary habitability.

10.5 Conclusion

The machine learning framework developed in this project successfully identifies patterns in planetary and stellar data that are strongly associated with potential habitability. By combining feature engineering, supervised learning models, and interpretability techniques, the system provides a scalable approach for analyzing large exoplanet datasets. The final output — a ranked list of potentially habitable exoplanets — offers valuable insights that may support future astronomical research and exoplanet discovery missions.